Assignment #1

Due Date: January 16 2026

Objectives

- Sorting Algorithms.
- Analyzing computational complexity using asymptotic notation.
- Practicing writing clear and rigorous solutions that communicate reasoning effectively.

Submission Instructions

Solve the following problems. Write your solutions on a paper or type them out. Submit your solution to the correspondin dropbox folder. Submit the file as a pdf file.

- Name your file: a1_x.pdf where x is you username as it appears on MyLearningSpace. For example, if I were submitting the assignment, I would name my file: a1_mrudafshani.pdf.
- Write neatly and clearly, or you may lose marks.
- Include the following at the beginning of your submission.

I certify that this assignment is entirely my own work and that all external resources, if used, have been properly acknowledged.

Problem 1

Analyze the time complexity of the following code snippets and express it using Θ notation.

(a)
```
for(i = 1; i <= n; i = i*2) {
    for(j = 1; j <= i; j++)
        print (j);
}
```

Hint: Please note: When n = 1, internal loop runs once
When n = 2, internal loop runs 1+2 times
When n = 3, internal loop runs 1+2 times
When n = 4, internal loop runs 1+2+4 times
When n = 5, internal loop runs 1+2+4 times
When n = 8, internal loop runs 1+2+4+8 times
When n = n, internal loop runs for 1+2+4+8+ ... ? times?

(b)
```
sum = 0;
for(i = n; i > 0; i = i/6)
    sum = sum+i;
```

(c)
```
sum = 0;
for(i = 1; i <= n; i = i++)
    for(j = 1; j < i; j = j*3)
      sum = sum+i;
```

Problem 2

(a) Can insertion sort be implemented on linked lists with the same $O(n^2)$ time complexity as its array-based version?

(b) You are given the results of a chess tournament with n players, where every pair of players played exactly once. Each match ended in a win for one player or a draw.

Design an algorithm that orders the players so that no player is listed immediately before a player they lost to. Explain the algorithm, and provide the pseudocode.

Analyze the time complexity of your algorithm?

Problem 3

Prove the following expressions:

- $4^n \notin O(2^n)$. Prove your answer by using definition of big O.
- $a^{\log_b x} = x^{\log_b a}$. Deduce your answer by applying logarithm principles.
- $\lg(n!) \in \Theta(n \lg n)$

Problem 4

Rank the following functions by their order of growth. Arrange them as $g_1, g_2, \ldots, g_{15}$ such that $g_1 = \Omega(g_2), g_2 = \Omega(g_3), \ldots, g_{14} = \Omega(g_{15})$.

$$\frac{n}{100} \qquad 2^{\lg n} \qquad n^2 \qquad n! \qquad (\lg n)!$$

$$\ln \ln n \qquad n^{100} \qquad n^{\lg \lg n} \qquad \ln n \qquad 1$$

$$2^{\lg n} \qquad (\lg n)^{\lg n} \qquad 4^{\lg n} \qquad (n+1)! \qquad \sqrt{\lg n}$$